

# Michael Xu

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## EDUCATION

### Washington University in St. Louis

St. Louis, MO

*M.S. in Computer Science, Minor in HCI, Minor in EE | GPA: 3.82*

*Aug. 2023 – May 2027*

- **Coursework:** DSA, Analysis of Algorithms, Systems, Parallel & Concurrent Programming, Web Dev

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, C#, JavaScript, SQL, C++, HTML/CSS

**Frameworks:** Spring Boot, React, Next.js, FastAPI

**Technologies:** Claude Code, AWS, PostgreSQL, Redis, Docker, Git, Linux, Supabase, Cursor

## EXPERIENCE

### Software Engineer Intern

June 2026 – Aug. 2026

*Zocdoc — React, TypeScript, C#*

*Soho, NY*

- Shipped a **React** payment-recovery module that unifies failed-payment status, balance, spend-lock state, and retry timing on the billing page, replacing a manual “update payment” flow behind **~20%** of billing support tickets
- Built a new **C#** payment endpoint that charges outstanding balances on demand outside the daily billing queue, cutting balance-resolution from **24h** to **~5 min** with locking and state management to prevent double-charges
- Extended an existing **C#** read endpoint to power the decline reasons and per-account restriction state, driving the retry-timing and unlock messaging — targeting a **50%** monthly cut in payment-confusion tickets (**~\$2.6M** at-risk)

### Software Engineer Intern

May 2025 - Sept. 2025

*National Information Solutions Cooperative (NISC) — Java, Spring, SQL*

*Lake St. Louis, MO*

- Re-architected legacy billing hooks into modular **Spring Boot** services with structured **Java** DAO/BO layers, replacing shell-script job orchestration and delivering a **40%** runtime speedup on **100M+** statement batches
- Partnered with **6 senior engineers** to deploy and validate billing modules on **bctest12**, review and merge PRs in **Bitbucket**, and coordinate directly with utility clients to secure test datasets and configuration files.
- Established a repeatable **PL/SQL-to-Java modernization pipeline** and migration playbook covering **100+** legacy billing hooks, accelerating onboarding and de-risking large-scale production releases.

### Software Engineer

Aug. 2025 - Present

*Ursaworks Robotics — React, Typescript, Javascript, Python*

*St. Louis, MO*

- Serve as sole frontend maintainer for a **React** platform supporting **100+** active members, shipping feature updates and bug fixes across all engineering teams
- Improved page load performance via component refactoring, asset optimization, and rendering flow restructuring, reducing load time by **40%** across high-traffic pages
- Delivered real-time autonomous perception for competition robots via a **YOLO-based** detection pipeline on **NVIDIA** Jetson hardware, contributing to a **top-8 of 24** finish at **RoboMaster** Nationals

## PROJECTS

### Workout Coach | React, Next.js, TypeScript, Supabase, YOLOv8

Mar. 2026 – Present

- **Finalist** at **HackWashU 2026** and **Mayfield AI Garage Pitch** — selected from **50+** competing teams
- Delivered real-time rep counting and live form feedback by integrating a **YOLOv8** pose estimation model into a **Next.js** frontend, enabling frame-by-frame workout quality analysis without backend roundtrips
- Accelerated AI response quality via configurable prompt tuning and multi-persona LLM routing, reducing irrelevant workout suggestions and improving session personalization
- Persisted user workout history and session state via a normalized **Supabase** schema, deployed end-to-end on **Vercel** with zero-config CI

### Musical Collaboration Network Analysis | Python, NetworkX, Spotify API

Nov. 2025 – Dec. 2025

- Parsed a **10GB+** Discogs XML dump via memory-efficient iterative parsing to build a music collaboration graph of **1,000+** artists and **10,000+** edges spanning **130 years** (1897–2025), enriched with **Spotify API** genre data
- Predicted future collaborations using graph link-prediction methods (CN, Jaccard, AA, Louvain) in **NetworkX**, reaching **0.85 AUC** via a grid-search-optimized hybrid model combining structural and community features
- Built an era-based temporal framework splitting the network into **5** musical periods with chronological train/test splits, outperforming a single global model by **5–10% AUC**